Actividad 3 - Interpolaciones

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Primer punto:

Nota: Utilice interpolaciones cuadráticas en todos los incisos.

Formula de la interpolación cuadrática:

$$\boldsymbol{f(x) = f(x_0)\frac{(x-x_1)(x-x_2)}{(x_0-x_1)(x_0-x_2)} + f(x_1)\frac{(x-x_0)(x-x_2)}{(x_1-x_0)(x_1-x_2)} + f(x_2)\frac{(x-x_0)(x-x_1)}{(x_2-x_0)(x_2-x_1)}}$$

a) La densidad es de 50ºC y 28% de concentración.

$$A\,T = 40°C, densidad = 1.2652$$
$$A\,T = 80°C, densidad = 1.2418$$
$$A\,T = 100°C, densidad = 1.2301$$

$$x_0 = 40, f(x_0) = 1.2652$$
$$x_1 = 80, f(x_1) = 1.2418$$
$$x_2 = 100, f(x_2) = 1.2301$$
$$x = 50$$

Resultado: La densidad a 50°C y 28% de concentración es aproximadamente 1.2594

b) La densidad es de 90ºC y 25% de concentración.

$$A\,T = 80°C, las\ densidades\ son\ 1.1570\ y\ 1.2418$$
$$AT = 100°C, las\ densidades\ son\ 1.1451\ y\ 1.2301$$

$$Interpolar\ la\ densidad\ para\ la\ concetracion\ C = 25\%\ entre\ C = 20\%\ y\ C = 28\%\ usando\ la\ interpolacion\ lineal$$
$$Densidad\ a\ C = 25\%\ y\ T = 80°C = 1.2041$$
$$Densidad\ a\ C = 25\%\ y\ T = 100°C = 1.1876$$

$$Interpolar\ la\ temperatura\ entre\ T = 80°C\ y\ T = 100°C\ para\ C = 25\%:$$
$$f(90) = 1.2041$$
$$Resultado: La\ densidad\ a\ 90°C\ y\ 25\%\ de\ concentracion\ es\ aproximadamente\ 1.2041$$

c) La densidad es de 40ºC y 15% de concentración.

$$Para\ concentraciones\ C = 12\%\ y\ C = 20\%\ a\ T = 40°C:$$
$$f(C = 15\%) = 1.0521$$
$$Resultado: La\ densidad\ a\ 40°C\ y\ 15\%\ es\ aproximadamente\ 1.0521$$

d) d. La concentración que tiene una solución de densidad 1.129 a una temperatura de 60ºC.

$$T = 60°C \quad Datos: 40°C\ y\ 80°C$$

$$C = 4\%: 1.0276\ a\ 40°C\ y\ 1.0063\ a\ 80°C$$
$$C = 12\%: 1.1013\ a\ 40°C\ y\ 1.0786\ a\ 80°C$$
$$C = 20\%: 1.1801\ a\ 40°C\ y\ 1.1570\ a\ 80°C$$
$$C = 28\%: 1.2652\ a\ 40°C\ y\ 1.2418\ a\ 80°C$$

Los resultados interpolados en 60°C son

$$C = 4\%, f(60°C) = 1.0170$$
$$C = 12\%, f(60°C) = 1.0900$$
$$C = 20\%, f(60°C) = 1.1686$$
$$C = 28\%, f(60°C) = 1.2535$$

Interpolar la concentración para la densidad f= 1.129 y las concentraciones de C = 12% y C= 20%

$$C = 15.97\%$$
$$Resultado: La\ concentracion\ para\ una\ densidad\ de\ 1.129\ a\ 60°C\ es\ de\ 15.97\%$$

Segundo punto:

Datos:
$T = 150°C\ y\ T = 200°C$
$Presiones: P = 4, P = 7\ y\ P = 10$

Polinomio para T = 150°C
$A\ P = 4, V = 38.14$
$A\ P = 7, V = 46.29$
$A\ P = 10, V = 56.48$

$a_0 = 5.7094$
$a_1 = -10.5133$
$a_2 = 4.6906$

El polinomio para T = 150°C es:

$V_{150}(P) = 4.6906P^2 - 10.5133P + 5.7094$

Polinomio para T = 200°C
$A\ P = 4, V = 69.38$
$A\ P = 7, V = 89.18$
$A\ P = 10, V = 118.32$

$a_0 = 11.5278$
$a_1 = -20.8556$
$a_2 = 8.8089$

El polinomio para T = 200°C es:

$V_{200}(P) = 8.8089P^2 - 20.8556P + 11.5278$

$T = 150°C$:
$V_{150}(7) = 4.4906 * 7^2 - 10.5133 * 7 + 5.7094$
$V_{150}(7) = 46.29$

$T = 200°C$:
$V_{200}(7) = 8.8089 * 7^2 - 20.5133 * 7 + 11.5278$
$V_{200}(7) = 89.18$

Fórmula para la interpolación lineal es:

$$V(T) = V(T_1) + \big(V(T_2) - V(T_1)\big) * \frac{T - T_1}{T_2 - T_1}$$

Datos:

$T_1 = 150°C$
$T_2 = 200°C$
$T = 175°C$
$V(T_1) = 46.29$
$V(T_2) = 89.18$

$V(175) = 46.29 + (89.18 - 46.29) * \frac{175 - 150}{200 - 150}$
$V(175) = 46.29 + 42.89 * 0.5$
$V(175) = 46.29 + 21.445 = 67.735$

Resultado:
$T = 175°C\ y\ P = 7\ es\ de\ 67.74\ moles/litro$

Tercer punto:

$$a_2 = \frac{\frac{f[x_2]-f[x_1]}{x_2-x_1} - \frac{f[x_1]-f[x_0]}{x_1-x_0}}{x_2-x_0} = f[x_0,x_1,x_2]$$

$$\frac{f[x_2]-f[x_1]}{x_2-x_1} - \frac{f[x_1]-f[x_0]}{x_1-x_0}$$

$$\frac{(f[x_2]-f[x_1])(x_1-x_0)-(f[x_1]-f[x_0])(x_2-x_1)}{(x_2-x_1)(x_1-x_0)}$$

$$\frac{\frac{(f[x_2]-f[x_1])(x_1-x_0)-(f[x_1]-f[x_0])(x_2-x_1)}{(x_2-x_1)(x_1-x_0)}}{x_2-x_0}$$

$$\frac{(f[x_2]-f[x_1])(x_1-x_0)-(f[x_1]-f[x_0])(x_2-x_1)}{(x_2-x_1)(x_1-x_0)(x_2-x_0)}$$

$$a_2 = f[x_0,x_1,x_2]$$

Cuarto punto:

$$f[\,x_i, x_i+1\,] = \frac{f(x_i+1)-f(x_i)}{x_i+1-x_i}$$

$$f[\,40{,}60\,] = \frac{1.36-0.63}{60-40} = \frac{0.73}{20} = 0.0365$$

$$f[\,60{,}80\,] = \frac{2.18-1.36}{80-60} = \frac{0.82}{20} = 0.041$$

$$f[\,80{,}100\,] = \frac{3.00-2.18}{100-80} = \frac{0.82}{20} = 0.041$$

$$f[\,100{,}120\,] = \frac{3.93-3.00}{120-100} = \frac{0.93}{20} = 0.0465$$

$$f[\,120{,}140\,] = \frac{6.22-3.93}{140-120} = \frac{2.29}{20} = 0.1145$$

$$f[\,140{,}160\,] = \frac{8.59-6.22}{160-140} = \frac{2.37}{20} = 0.1185$$

$$f[\,40,60,80\,] = \frac{0.041 - 0.0365}{80 - 40} = \frac{0.0045}{40} = 0.0001125$$

$$f[\,60,80,100\,] = \frac{0.041 - 0.041}{100 - 60} = 0$$

$$f[\,80,100,120\,] = \frac{0.0465 - 0.041}{120 - 80} = \frac{0.0055}{40} = 0.0001375$$

$$f[\,100,120,140\,] = \frac{0.1145 - 0.0465}{140 - 100} = \frac{0.068}{40} = 0.0017$$

$$f[\,120,140,160\,] = \frac{0.1185 - 0.1145}{160 - 120} = \frac{0.004}{40} = 0.0001$$

$\boldsymbol{e}$	$\boldsymbol{p}$	$\boldsymbol{f[\,x_0, x_1\,]}$	$\boldsymbol{f[\,x_0, x_1, x_2\,]}$
40	0.63		
60	1.36	0.0365	
80	2.18	0.0410	0.0001125
100	3.00	0.0410	0
120	3.93	0.0465	0.0001375
140	6.22	0.1145	0.0017
460	8.59	0.1185	0.0001

b.

$$P(x) = f[x_0] + f[x_0, x_1](x - x_0) + f[x_0, x_1, x_2](x - x_0)(x - x_1)$$

$$x_0 = 80\,, x_1 = 60, x_2 = 40$$

$$f[40] = 0.63, f[40,60] = 0.0365, f[40,60,80] = 0.0001125$$

$$P(90) = 0.63 + 0.0365(90 - 40) + 0.0001125(90 - 40)(90 - 60)$$

$$0.0365(90 - 40) = 0.0365(50) = 1.825$$

$$0.0001125(90 - 40)(90 - 60) = 0.0001125(50)(30) = 0.16875$$

$$P(90) = 0.63 + 1.825 + 0.16875 = 2.62375$$

Quinto punto:

$x_2 = 8, \qquad x_3 = 11.5, \qquad x_4 = 15$
$f(8) = 27500$
$f(11.4) = 1900$
$f(15) = 15700$

$$f[x_2, x_3] = \frac{f(11.5) - f(8)}{11.5 - 8} = \frac{19000 - 27500}{11.5 - 8} = \frac{-8500}{3.5} = -2428.57$$

$$f[x_3, x_4] = \frac{f(15) - f(11.5)}{15 - 11.5} = \frac{15700 - 19000}{15 - 11.5} = \frac{-3300}{3.5} = -942.86$$

$$f[x_2, x_3, x_4] = \frac{-942.86 - (-2428.57)}{15 - 8} = \frac{1485.71}{7} = 212.24$$

$$P(x) = f(x_2) + f[x_2, x_3](x - x_2) + f[x_2, x_3, x_4](x - x_2)(x - x_3)$$

$$P(x) = 27500 + (-2428.57)(x - 8) + 212.24(x - 8)(x - 11.5)$$

$-2428.57(10 - 8) = -2428.57 * 2 = -4857.14$
$212.24(10 - 8)(10 - 11.5) = 212.24 * 2 * (-1.5) = -636.72$
$P(10) = 27500 - 4857.14 - 636.72 = 22006.14$

Estimación del error:

$$E(10) = \frac{212.24}{6}(10 - 8)(10 - 11.5)(10 - 15)$$

$$10 - 8 = 2$$
$$10 - 11.5 = -1.5$$
$$10 - 15 = -5$$
$$2 * -1.5 * -5 = 15$$

$$E(10) = \frac{212.24}{6} * 15 = 35.3733 * 15 = 530.60$$

$$x = 10 \rightarrow \quad 22006.14 \pm 530.60$$